INTERNATIONAL ISLAMIC UNIVERSITY CHITTAGONG

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

Project Report On :

Course Code : CSE-3644
Course Title : Software Development II Lab

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1. Introduction

The objective of this project was to develop a personal portfolio application using Flutter while applying the concepts learned throughout the SD2 Lab course. The project required implementing size and color transition animations on at least three different user interface components and organizing the application using the GetX architecture. This assignment provided an opportunity to transform theoretical knowledge into a practical mobile application and strengthen our Flutter development skills.

2. Project Overview

The portfolio application is designed to present personal and professional information in an attractive and interactive way. It includes multiple sections such as personal details, educational background, technical skills, projects, achievements, and contact information. The application emphasizes a clean user interface, smooth navigation, and responsive design to provide a better user experience.

3. Technologies Used

The following technologies and tools were used to develop the project:

- **Flutter** – Used as the main framework for developing the cross-platform mobile application.
- **Dart** – The programming language used for Flutter application development.
- **GetX** – Used for state management, dependency injection, and route management.
- **Flutter Animation Framework** – Used to implement size and color transition animations.
- **Visual Studio Code / Android Studio** – Used as the development environment.
- **Git & GitHub (Optional)** – Used for version control and project management.

4. Implementation of Assignment Requirements

4.1 Size and Color Transition Animations

One of the major requirements of the assignment was implementing animations. In this project, size and color transition animations were applied to at least three different UI components. These animations improve the application's appearance and create a more interactive experience for users. They also demonstrate our understanding of Flutter's built-in animation capabilities.

4.2 GetX Architecture

The application follows the GetX architecture for organizing the codebase. GetX was used for state management and navigation, allowing the business logic to remain separate from the user interface. This architecture improves code readability, maintainability, and scalability while reducing unnecessary boilerplate code.

5. Reflection on the SD2 Lab Course

This project reflects the practical knowledge gained throughout the SD2 Lab course. During the semester, we learned the fundamentals of Flutter, including widgets, layouts, navigation, and responsive user interface design. These concepts formed the foundation of the portfolio application.

The course also introduced GetX, which simplified state management and application architecture. By applying GetX in this project, we gained practical experience in organizing Flutter applications efficiently.

Additionally, we learned the basics of API integration, including sending HTTP requests and displaying data from external sources. Although this project did not require Firebase or advanced backend development, the knowledge gained about Flutter and API integration provides a strong foundation for developing more complex applications in the future.

Building this project also improved our debugging, problem-solving, and project organization skills. We learned how to create reusable widgets, maintain clean code, and implement interactive features that improve the user experience.

6. Challenges Faced

During development, several challenges were encountered, including implementing smooth animations, maintaining proper widget organization, and managing application states efficiently using GetX. Debugging layout issues and ensuring responsive design across different screen sizes also required careful testing and refinement. Overcoming these challenges significantly improved our understanding of Flutter development.

7. Learning Outcomes

Through this project, we achieved the following learning outcomes:

- Developed a complete mobile application using Flutter.
- Applied GetX architecture for state management and navigation.
- Implemented size and color transition animations for multiple UI components.
- Improved understanding of Flutter widgets, layouts, and responsive design.
- Strengthened debugging and problem-solving skills.
- Gained practical experience in organizing a real-world Flutter project.

8. Conclusion

The Flutter Portfolio Application successfully fulfills the assignment requirements while demonstrating the concepts learned in the SD2 Lab course. It integrates Flutter fundamentals, GetX architecture, and animation techniques into a well-structured mobile application. The project provided valuable hands-on experience and reinforced the theoretical concepts taught during the course. Overall, it has strengthened our confidence in Flutter development and established a solid foundation for building more advanced mobile applications in future academic and professional projects.